

CANDIDATE NAME: _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2017

BIOLOGY
Higher 2
9744

ANSWER SCHEME

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in all sections.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Paper 1 (MCQ)	/30
Paper 2	
1	/15
2	/20
3	/15
4	/15
5	/18
6	/17
P2 Total	/100
Paper 3	/75
Paper 4	/55
TOTAL (100%)	

This question paper consists of 19 printed pages.

Answer **all questions** in the spaces provided.

Question 1

There have been many breakthroughs in stem cell research in recent years. It has been discovered that stem cells are involved in the replacement of worn-out cells and repair of damaged tissues. Further research is being conducted to better understand the mechanism involved in controlling the behaviour of stem cells in order to better manipulate them to treat various diseases and disorders.

(a) State the type of stem cells involved in the replacement of worn-out cells and repair of damaged tissues, and describe the unique properties of this type of stem cells. [2]

- **Adult stem cells** [1]

Any 2 properties [1]:

- **Undifferentiated cells found in differentiated tissues**
- **Multipotent → Able to differentiate into a limited range of cell types**
- **Able to undergo mitotic cell division for self-renewal**

Stem cells undergo cell division to produce genetically identical daughter cells. **Fig. 1.1** shows two cells, each at a different stage of cell division.

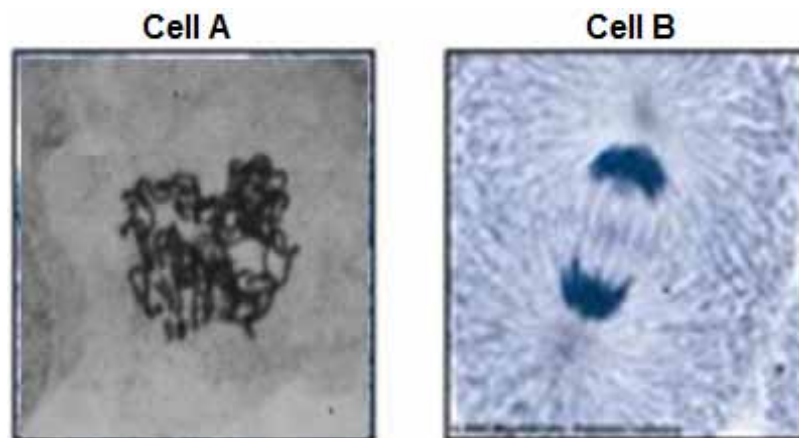


Fig. 1.1

(b)(i) With reference to **Fig. 1.1**, state the stages of cell division in **Cell A** and **Cell B**. [1]

Cell A: Prophase

Cell B: Anaphase

- (ii) The dysregulation of cell cycle can result in cancer. Outline the checkpoints that are present in normal cells to prevent this from occurring. [2]

Any 2

- **G₁ checkpoint:** assesses if the environmental conditions (presence of growth factors and nutrients, absence of DNA damage, adequate cell size) are favourable for cell division to proceed
- **G₂ checkpoint:** assesses if DNA replication is completed and cell size is adequate.
- **M checkpoint:** assesses if all chromosomes are attached to the mitotic spindle at their kinetochores and arrests the mitotic cell at metaphase if centromeres are not properly attached to kinetochore microtubules, hence preventing entry into anaphase.

Fig. 1.2 shows information about the movement of chromatids in a cell that has just started metaphase of mitosis.

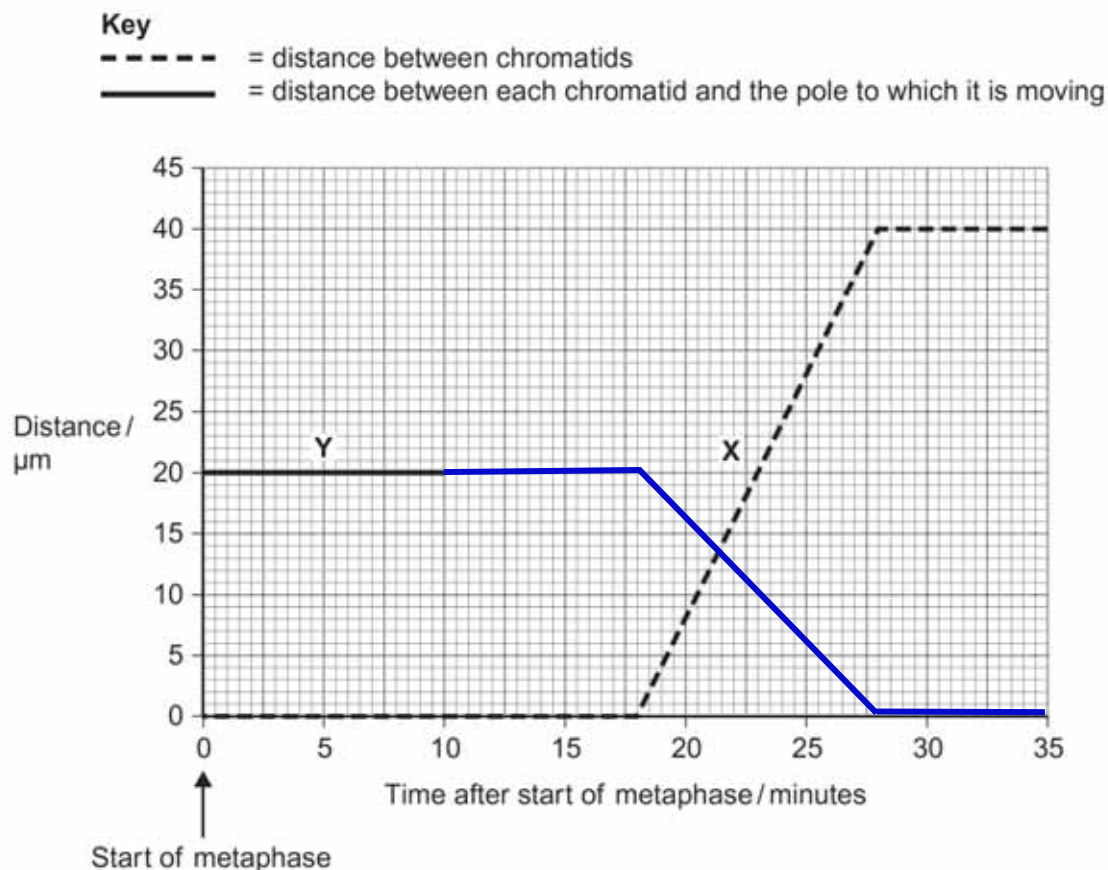


Fig. 1.2

- (c)(i) With reference to Fig. 1.2, state the duration of metaphase in the cell. [1]

- 18 min

- (ii) Complete line Y on the graph. [1]
- Horizontal until 18 minutes, then decreases as straight line to 0 μm at 28 minutes

- (iii) Account for your answer in (c)(ii). [3]

- Chromosomes align singly at the metaphase plate during metaphase of mitosis OR sister chromatids are attached to microtubules from opposite poles at metaphase
- Sister chromatids start to separate to become daughter chromosomes and migrate towards the opposite poles in anaphase, as shown at 18th min of line X when distance between chromatids starts to increase. Hence distance between chromatid and pole will start to decrease at 18th min.
- Distance between chromatids reach a plateau/maximum at 28th min, chromosomes arrived at opposite poles. Hence, distance between chromatid and pole will be minimum at 28th min.

The movement of chromatids is dependent on spindle fibres, which are made up of many tubulin subunits. Spindle fibres are lengthened at one end during mitosis by the polymerisation of tubulin subunits through GTP hydrolysis.

A drug, eribulin, is known to prevent the polymerisation of the tubulin subunits.

- (d)(i) Contrast between the structure of tubulin with that of DNA. [2]

Tubulin	DNA
• Tubulin is a polypeptide and hence made up of amino acid subunits	• DNA is made up of deoxyribonucleotides.
• Subunits by peptide bonds	• Subunits joined by phosphodiester bonds
• Globular	• Helical

- (ii) Suggest how eribulin work to prevent tubulin polymerisation and explain its effect on the behaviour of chromosomes in mitosis. [3]

- Possible action of drug: bind to tubulin subunits to change its conformation that prevent joining of subsequent subunits
- Prevent kinetochore microtubules from attaching to the kinetochores at the centromeres of the chromosomes.
 - Cells cannot progress through metaphase, so that chromosomes cannot align singly at the metaphase plate OR sister chromatids could not separate/remain attached.

[Total: 15]

Question 2

The retrovirus, human immunodeficiency virus (HIV), and the influenza virus are two types of enveloped viruses. Both enter the human host cells by adsorption and penetration. **Fig. 2.1** shows the entry process of a HIV into a macrophage, which is a type of white blood cell.

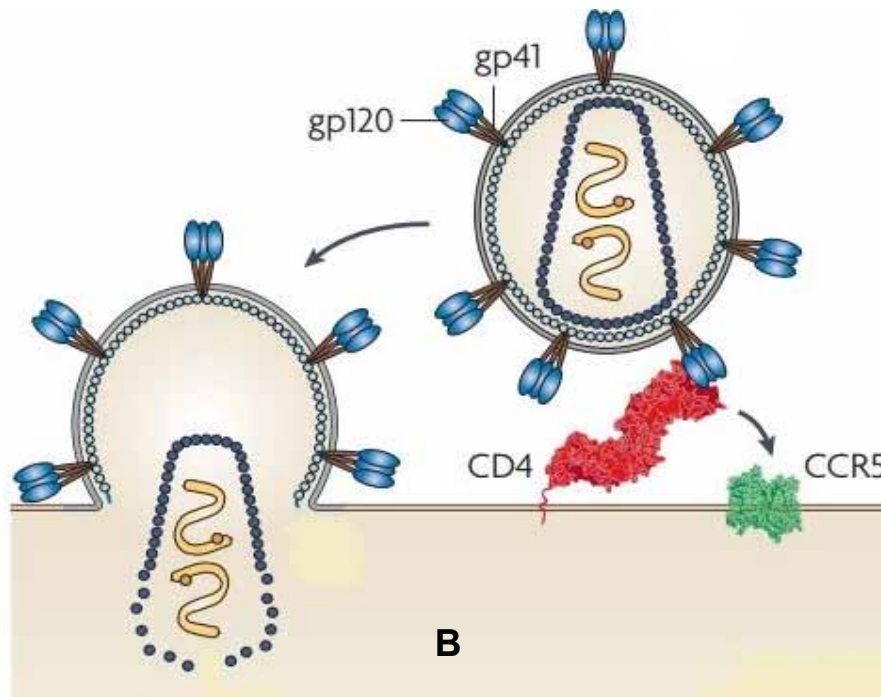


Fig. 2.1

(a)(i) State what is meant by *retrovirus*.

[2]

- Viruses with single stranded RNA as genome
- Has its own reverse transcriptase to produce DNA from its RNA genome

(ii) Compare the entry processes of the HIV and influenza virus into human host cells. [3]

- **Similarity:** Adsorption/ attachment of both viruses are by binding to specific cell surface receptors; AVP
- **Differences (any 2) :** Glycoprotein gp120 on the surface of the HIV binds to CD4, a cell-surface receptor found on white blood cells/ T helper cells / macrophages of the host immune system ; while haemagglutinin on the influenza viral membrane binds to sialic acid-containing receptors on the host cell membrane ;
 - The whole influenza virus enters the host cell by receptor-mediated endocytosis; while only the HIV capsid enters via membrane fusion where its envelope fuses with the host cell membrane;
 - Upon entry, the influenza virus forms an endosome / endocytic vesicle whereas the HIV virus does not form an endosome/ endocytic vesicle, the HIV releases the viral contents into the host cell cytoplasm.

(iii) Upon completion of the entry process, describe how the HIV genome is inherited by macrophage daughter cells. [3]

- RNA is reverse transcribed to complementary DNA strand by the enzyme reverse transcriptase.
- The enzyme integrase catalyses the integration of the viral DNA into the host chromosome which exist as a provirus.
- The provirus genome is also replicated along with the host cell genome and all daughter cells inherit the HIV genome

(iv) 'HIV-positive patients usually develop weak immunity.'

Explain what is meant by this statement. [4]

- Host cells of HIV viruses are macrophages and CD4 helper T cells
- Provirus begins viral replication to make viral proteins. Immune system responds by destroying the infected helper T cells.
- CD4 helper T cells level will decrease the ability of the patient to fight infections
- Patients develop AIDS and become more susceptible to opportunistic infections.

Another example of an enveloped virus is the herpes simplex virus. Viral DNA enters the nucleus of the host cell via nuclear pores and *directs synthesis of viral RNA and DNA*. The virus is able to grow in non-dividing cells because its genome encodes enzymes such as viral DNA polymerase and thymidine kinase. Thymidine kinase is involved in synthesising deoxyribonucleotides required for DNA replication.

(b)(i) Describe the normal function of nuclear pores. [2]

- Allows the movement of molecules across nuclear membrane from the nucleus to cytoplasm and vice versa
- Named example of molecules moving out of nucleus: mature mRNA / newly formed ribosome / tRNA OR into nucleus: enzymes involved in transcription/replication / ATP/ histones / spliceosome proteins

(ii) State the process that is involved in the [1]

Synthesis of viral RNA: **transcription**

Synthesis of viral DNA: **replication**

(iii) Describe two ways in which the named processes in (ii) are different. [2]

Replication	Transcription
Product is DNA molecule	Product is mRNA
Enzyme involved is DNA polymerase	Enzyme involved is RNA polymerase
Both strands of DNA as template	Only one strand serves as template

AVP

Acyclovir is an anti-viral drug which is used to treat herpes. It prevents the complete replication of viral DNA by viral DNA polymerase. The structure of acyclovir is shown below in **Fig. 2.2**.

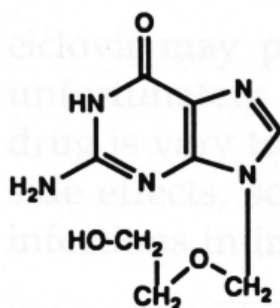


Fig. 2.2

When acyclovir enters the infected cell, it is first phosphorylated by thymidine kinase and subsequently becomes further phosphorylated by host cell kinases. When this phosphorylated acyclovir is incorporated into the newly-synthesised DNA strand, it prevents further elongation of the DNA strand.

(c)(i) With reference to **Fig. 2.2**, suggest how the incorporation of the phosphorylated acyclovir prevents further elongation of the DNA strand. [2]

- Missing 3'OH group
- DNA polymerase cannot recognise/bind to 3' end of DNA strand and cannot catalyse formation of phosphodiester bonds between DNA strand and incoming deoxyribonucleotide

(ii) Some strains of herpes simplex virus are now resistant to acyclovir. Suggest how the virus has gained resistance to this anti-viral drug. [1]

- Mutation that affects the phosphorylation of acyclovir by thymidine kinase or other host cell kinases

[Total: 20]

Question 3

In a particular variety of tomato plant, the allele for red fruit colour (**A**) is dominant over the allele for orange fruit (**a**) and the allele for green base when ripe (**B**) is dominant over the allele for no green base when ripe (**b**).

Two students, Faiz and Jacob crossed plant with red fruit and green base when ripe with pure bred plant with orange fruit and no green bases when ripe. The phenotypes of 50 offspring of each of Faiz's and Jacob's crosses were recorded and are shown in **Table 3.1**.

Table 3.1

	Phenotypes of offspring of test crosses			
	Red fruit with green base	Red fruit with no green base	Orange fruit with green base	Orange fruit with no green base
Faiz's cross	23	4	3	20
Jacob's cross	3	21	23	3

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (a)(i) With the aid of the table of probabilities as shown above, carry out a χ^2 test on the results of Faiz's cross and provide a brief explanation for the results obtained. [6]

Null Hypothesis: There is **no significant difference** between the **observed occurrence** of tomato plant phenotypes and the **expected occurrence ratio of 1:1:1:1**. [1]

Derivation of χ^2 calculated values: [1]

Category/Class	Observed Frequency (O)	Expected Frequency (E)	(O-E)	$\frac{(O-E)^2}{E}$
Red fruit with	23	$\frac{1}{4} \times 50 = 12.5$	10.5	8.82

green base				
Red fruit with no green base	4	$\frac{1}{4} \times 50 = 12.5$	-8.5	5.78
Orange with green base	3	$\frac{1}{4} \times 50 = 12.5$	-9.5	7.22
Orange with no green base	20	$\frac{1}{4} \times 100 = 25$	-5	4.50

Calculation of $\chi^2_{\text{calculated}}$ value

$$\chi^2_{\text{calculated}} = 8.82 + 5.78 + 7.22 + 4.50 = \mathbf{26.32 \text{ (3 s.f.)}}$$

[1]

Degree of freedom = 4 - 1 = 3

From the χ^2 table,

At 5 % level of significance and $\nu = 1$, $\chi^2_{\text{critical}} = \mathbf{7.82}$

Conclusion: Since $\chi^2_{\text{calculated}} > \chi^2_{\text{critical}}$ / $\mathbf{26.32 > 7.82}$, the **calculated χ^2 value** is higher than the **critical value at $p = 0.05$** , the **difference between the observed and expected results** is **statistically significant**. [1]

The difference between the observed and expected is not due to **chance alone**. Hence, **H_0 is rejected/not accepted** and the pattern of inheritance in the survey does not follow Mendelian's pattern of inheritance.

Explanation: A and B gene loci are linked/found on the same chromosome. [1] The genes do not undergo independent assortment and are usually inherited together. Hence, there is a higher proportion of the offspring having the parental phenotypes. [1]

- (ii) State the probability that the results of Faiz's cross depart significantly by chance from the expected ratio. [1]

$p < 0.001$

- (b) Draw a genetic diagram to explain the results of Jacob's cross. [5]

Parental phenotype: Red fruit, Green base x Orange fruit, no green base

Parental genotype: $\frac{A b}{a B}$ x $\frac{a b}{a b}$

Gametes: $\frac{A b}{A B}$; $\frac{a B}{a b}$; x $\frac{a b}{a b}$;

F₂ genotypes

	$\frac{A b}{a b}$	$\frac{a B}{a b}$	$\frac{A B}{a b}$	$\frac{a b}{a b}$
$\frac{a b}{a b}$	$\frac{A b}{a b}$	$\frac{a B}{a b}$	$\frac{A B}{a b}$	$\frac{a b}{a b}$

F₁ phenotypes

F₁ phenotypic ratio

Red fruit with no green base	Orange fruit with green base	Red fruit with green base	Orange fruit with no green base
21	23	3	3

1m each

(c) Explain the difference in results between Faiz's and Jacob's cross when the parental genotypes are the same. [3]

- Parental genotypes of red with green base tomato plant used are the same however the combination of the alleles of the two gene loci on the same chromosome are different
- Show or explain the genotypes of both parent tomato plant in two crosses
- $\underline{A B} \times \underline{a b}$ for Faiz's cross;
 $a b \quad a b$
- $\underline{A b} \times \underline{a b}$ for Jacob's cross
 $a B \quad a b$
- Since the genes are linked, they are usually inherited together and hence would result in an overrepresentation of parental phenotypes among offsprings.

[Total: 15]

Question 4

Yeast haploid cells secrete α factor to signal mating, and respond by growing a mating projection towards a potential mate. Upon contact of the two partner cells, these fuse to form a diploid zygote.

Fig. 4.1 shows the α factor signaling pathway mediated by yeast G-protein coupled receptor, Ste2. The activation of the pathway induces the expression of *FUS1* gene which is required for yeast mating.

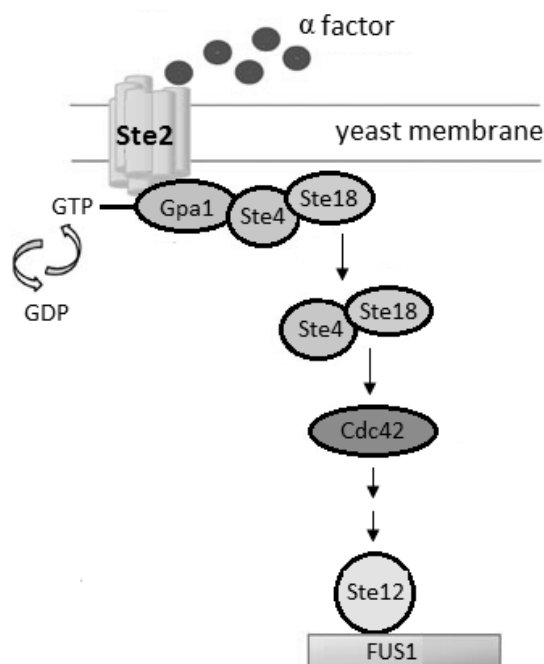


Fig. 4.1

(a)(i) With reference to **Fig. 4.1**, describe how α factor triggers the activation of Cdc42. [4]

- α factor binds to Ste2 receptor at the binding site and triggers a conformational change in Ste2
- allowing Ste2 to bind Gpa1(-Ste4-Ste18 complex) and causes GTP to displace GDP
- activation of the (Gpa1-)Ste4-Ste18 complex
- Ste4-Ste18 dimer to dissociate from the complex to activate Cdc42

(ii) Briefly explain why α factor cannot enter the yeast cell directly. [2]

- α factor is polar / charged or too large to pass through transient gaps
- cannot pass through the hydrophobic core of the phospholipid bilayer and requires a membrane receptor

(iii) Explain the possible role of Ste12 in the expression of FUS1 gene. [2]

- General transcription factor that binds to the promoter of FUS1 gene
- Allows for the assembly of the transcription initiation complex to initiate transcription

(iv) It has been observed that the binding of a protein Y to a region upstream of the promoter results in the mating projection to be produced at a rate higher than normal in the yeast cells.

Provide a reason for this observation. [3]

- Region upstream of promoter is an enhancer
- Protein Y acts as an activator to bind to region upstream of promoter to increase the rate of transcription by
- Recruit histone acetylases to loosen chromatin cycle and increase the rate of assembly of the transcription initiation complex.

Fig. 4.2 shows the Ste2 receptor on another yeast cell membrane.

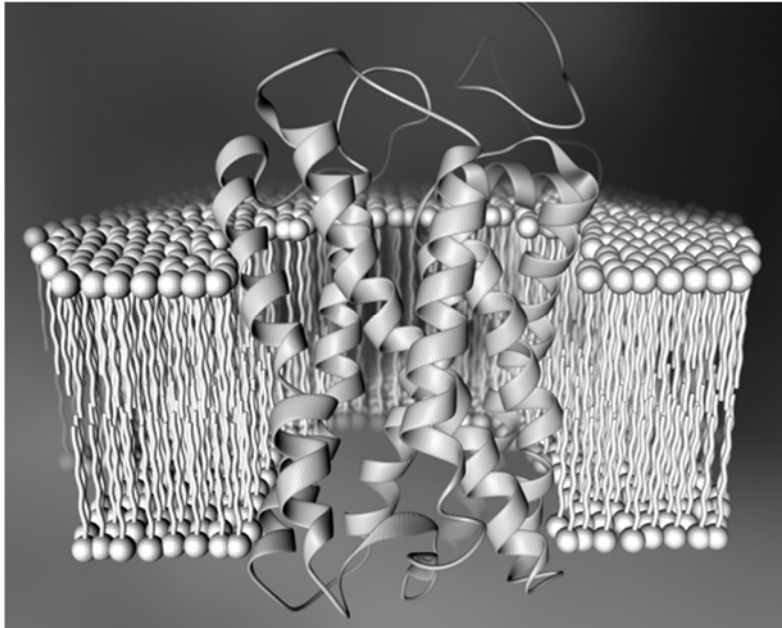


Fig. 4.2

- (b)(i) Describe the structure of Ste2 receptor. [2]
- A single polypeptide chain consisting of (seven) α helix segments that folds upon itself
 - Such that the α helices are gathered together forming a cylindrical / globular structure
- (ii) Explain how Ste2 receptor remains embedded in the yeast cell membrane. [2]
- Hydrophobic non polar R groups of amino acid residues that make up Ste2 receptor forms hydrophobic interactions with the hydrophobic fatty acid tails of the phospholipid.
 - Polar or charged R groups of amino acid residues that make up Ste2 receptor forms favorable interactions with the hydrophilic phosphate head of the phospholipid.

[Total: 15]

Question 5

Arabidopsis thaliana is a small flowering plant native to Asia. A mutation in the gene coding for NADP oxidase results in plants with short root hairs. NADP oxidase is an enzyme that converts NADPH to NADP⁺.

Fig. 5.1 shows the root hairs in the two variant of *A. thaliana*.

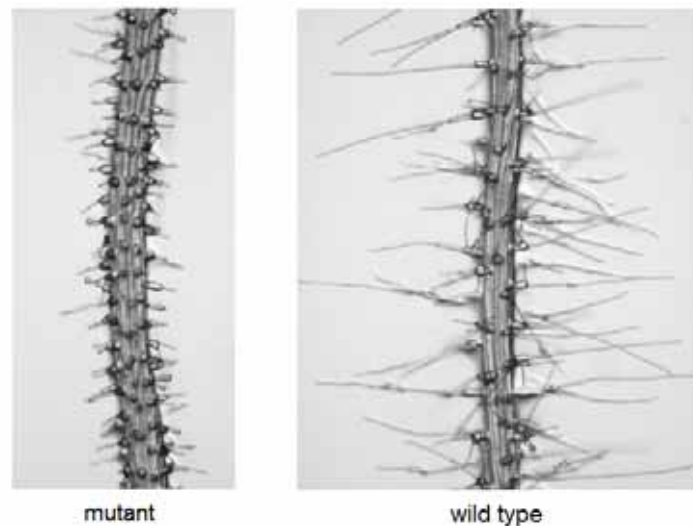


Fig. 5.1

(a) Explain the role of NADPH in photosynthesis.

[2]

- Provides reducing power/H⁺ to reduce
- Phosphoglyceric acid (PGA)/glycerate-3-phosphate (GP) to glyceraldehyde-3-phosphate (GALP)/phosphoglyceraldehyde (PGAL)/triose phosphate (TP)

In a separate experiment, activity of NADP oxidase in the tips of the root hair cells in wild type and mutant *A. thaliana* were measured at intervals. Changes in mean length of the root hair cell was also measured to track the rate of growth, which is known to be an energy-requiring process. Both sets of results are shown in **Fig. 5.2**.

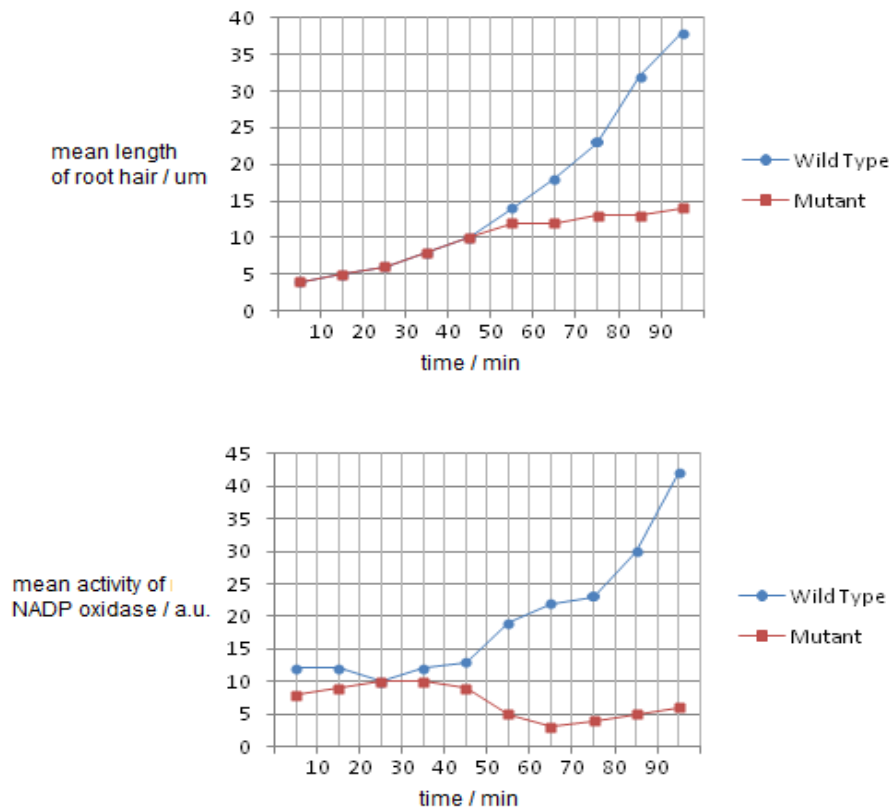


Fig. 5.2

(b) Using your knowledge of photosynthesis, provide an explanation for the difference in the growth of root hairs in the two types of *A. thaliana*. [4]

- Mean length of root hair for wild type *A. thaliana* increases from 4 μm to 38 μm when mean activity of NADP increases from 13 au to 42 au
- Mean length of root hair for mutant *A. thaliana* increases from 4 μm to 14 μm when mean activity of NADP decreases from 9 au to 6 au after a small initial increase
- NADP oxidase catalyses the conversion of NADPH to NADP⁺ for light reaction of photosynthesis where ATP is synthesised; (OR less reduction of PGA to GALP in Calvin cycle)
- Low NADP oxidase activity results in less carbohydrates produced and hence less growth

Rubisco is an enzyme required in the light-independent stage of photosynthesis. **Fig. 5.3** shows the effect of increasing temperature on the activity of two variations of Rubisco, **Rubisco C** and **Rubisco S**.

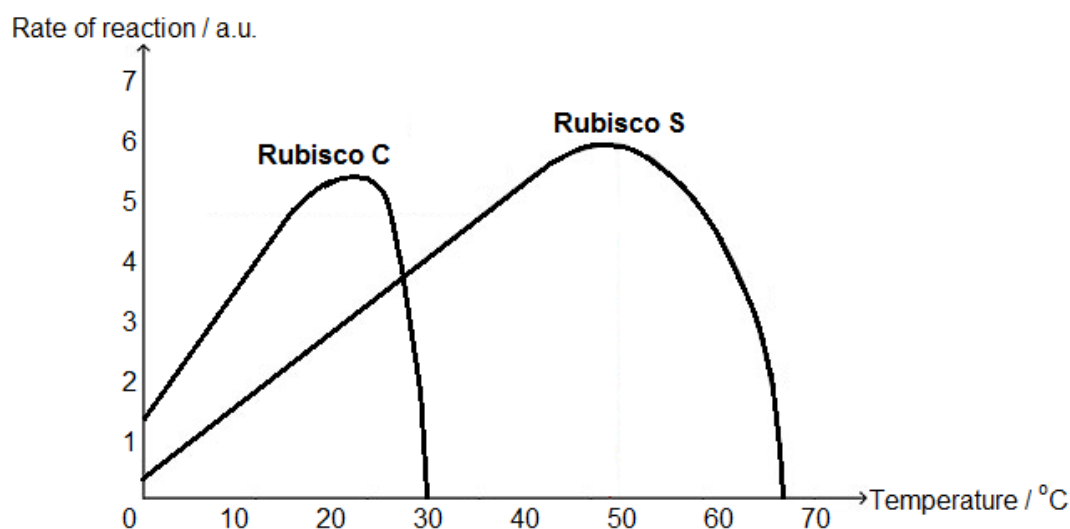


Fig. 5.3

(c) With reference to **Fig. 5.3**, compare the effect of temperature on the two enzymes. [3]

- Both Rubisco C and Rubisco S has an increased rate of reaction as temperature increases up to optimum temperature OR both Rubisco C and Rubisco S are denatured at temperatures higher than optimum.
- Rubisco C has a lower optimum temperature of 20°C as compared to Rubisco S at 50 °C where rate of reaction is at a maximum
- Rubisco C reaches a lower maximum rate of reaction of 5.5 a.u. at a faster rate as compared to Rubisco S which reaches a maximum rate of reaction of 6 a.u at a slower rate.

(d) It is known that **Rubisco C** is obtained from a species of coniferous tree found in Canada, while **Rubisco S** is obtained from a species of cactus found in the Sahara Desert.

(i) Explain how different alleles give rise to different Rubisco structure. [3]

- Different alleles have different DNA nucleotide sequence that results in a different mRNA/codon sequence after transcription
- Thus will result in different amino acid sequence / primary structure after translation
- Different R group interactions between amino acids affects folding of the polypeptide chain, giving rise to different 3D conformation in the tertiary structure

(ii) It has been predicted that **Rubisco S** will be found in more plant species in view of climate change. Explain how Darwin's theory of evolution supports this observation. [4]

- Variation: different types of Rubisco molecules that has different tolerance to high temperature
- Selection pressure: rising global temperatures that selects for plants with Rubisco S that are able to photosynthesise efficiently at high temperatures
- Plants with Rubisco S are able to survive and reproduce and pass down the favorable Rubisco S alleles to the next generation
- With time, allele frequency of the plant population gene pool will change with a higher frequency of the Rubisco S allele that confers a selective advantage, lower frequency of Rubisco C allele that confers selective disadvantage. Hence, more plants with Rubisco S.

(iii) Suggest two other ways plants can adapt to the changing climate. [2]

- Development of **more and longer roots** in order to access water deeper into the ground.
- **Reduction in the number and surface area of leaves** so as to lower transpiration rates.
- **Reduction in the number of stomata per leaf** to reduce transpiration rates.
- **Sunken stomata** which are concealed by **finger-like projections**.

[Total: 18]

Question 6

The table below shows the amino acid differences in the cytochrome b protein between various vertebrates.

	Human	Elephant	Platypus	Ostrich	Starling	Crocodile	Lungfish	Coelacanth	Goldfish	Shark
Human		26	40	43	41	47	83	70	68	71
Elephant			45	45	48	50	84	72	63	74
Platypus				54	52	51	89	74	70	76
Ostrich					26	36	91	75	68	73
Starling						47	91	77	67	70
Crocodile							85	78	70	77
Lungfish								90	94	86
Coelacanth									83	78
Goldfish										88
Shark										

Fig 6.1 shows the phylogenetic tree based on differences between the cytochrome b proteins.

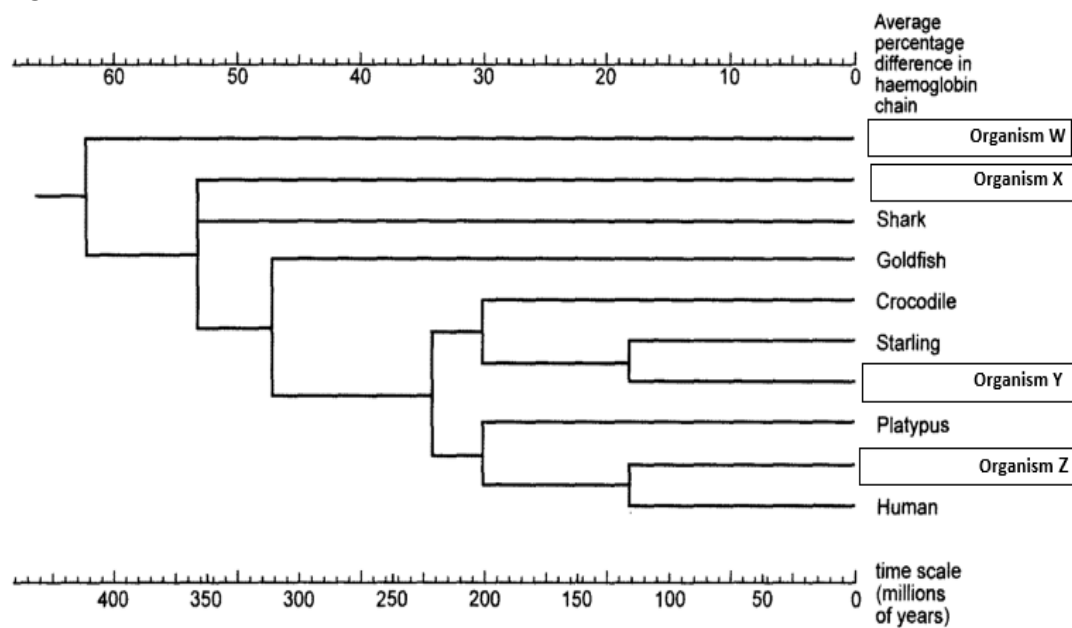


Fig. 6.1

(a) Using information from the table and **Fig. 6.1**, identify organisms **W** to **Z**. [2]

Organism **W**: **lungfish**

Organism **X**: **coelacanth**

Organism **Y**: **ostrich**

Organism **Z**: **elephant**

(b) Distinguish between classification and phylogeny. [2]

- **Classification refers to grouping organisms based on similar characteristics while phylogeny involves grouping organisms based on evolutionary relationship.**
- **Similar characteristics in classification may be analogous and not homologous whereas in phylogeny, similarity is due to inheritance from common ancestry.**

(c) Explain how differences in amino acid sequences in the cytochrome b chain allow the establishment of the phylogenetic tree. [2]

- **Percentage of amino acid difference indicates relatedness where few difference indicates recent common ancestor**
- **Provides quantitative data to construct phylogenetic tree**

(d) Suggest why homology still features prominently in evolutionary studies despite the advantages that molecular evidence can confer. [1]

- **Less expensive as it does not rely on machines**
- **DNA/protein samples might be limited or unavailable**

Brown adipocytes were one of the cells isolated for the above investigation. **Fig. 6.2** shows the schematic representation of a series of protein complexes found on the inner membrane of organelle **X** present in brown adipocytes.

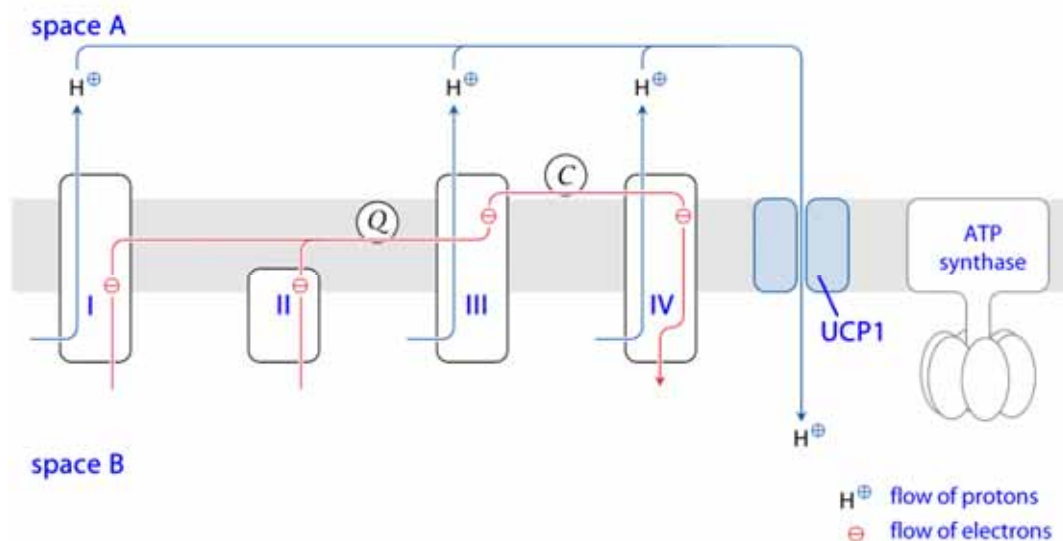


Fig. 6.2

(e)(i) State the identity of organelle **X**. [1]

Mitochondrion

(ii) Outline how ATP is usually synthesised in the inner membrane of organelle **X**. [4]

- **NADH and FADH₂ carry hydrogen in the form of protons and electrons where proton remain in the matrix and electrons are passed along the progressively lower energy electron carriers in the electron transport chain.**
- **Energy released is used to pump H⁺ from the matrix to the intermembrane space via active transport which sets up a concentration gradient (high H⁺ conc in intermembrane space, low conc in the matrix)**
- **H⁺ diffuse down the concentration gradient from the intermembrane space to matrix via the stalked particle**
- **Provides a proton motive force that drives the synthesis of ATP by ATP synthase by phosphorylation of ADP and inorganic phosphate (chemiosmosis).**

(f) Brown adipocytes contain a unique protein, UCP1, which is not found in organelle **X** in any other cell type.

Evaluate the impact of UCP1 on ATP synthesis and suggest the physiological significance of brown adipose tissue. [3]

- **As UCP1 allows protons to leak back into the matrix without passing through the ATP synthase,**
- **Loss of H⁺ concentration gradient, no ATP will be synthesized**
- **The energy released from the spontaneous flow of protons through UCP1 is lost as heat, which helps to keep the organisms warm.**

(g) In other cell types, NADH and FADH₂ are used to drive ATP synthesis by ATP synthase. Using relevant information from **Fig. 6.2**, suggest and explain why more ATP is produced from NADH. [2]

- **NADH donates electrons to complex I while FADH₂ donates to complex II. The energy released from transfer of electrons through the complexes is used to pump protons across the inner membrane.**
- **NADH allows for more chances to pumps more protons across the gradient, which powers the ATP synthase and gives us 3 ATP per molecule of NADH, while FADH₂ produces 2 ATP during the ETC because it gives up its electron to complex II, bypassing complex I.**

[Total: 17]

END OF PAPER